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# Midterm #1 Physics 135B

Unit 8:

① a) force on  $x=8\text{cm}$  if  $q=1.00\mu\text{C}$ ?  $+q=3\text{cm}$   $-2q=8\text{cm}$   $+q=11\text{cm}$

$$F = k \frac{q_1 q_2}{r^2}$$

$$+q = 3\text{cm}$$

$$r = 8 - 3 = 5 = 0.05\text{m}$$

$$F_1 = k \frac{(1 \times 10^{-6})(-2 \times 10^{-6})}{(0.05)^2}$$

$$= (9 \times 10^9) \frac{2 \times 10^{-12}}{0.0025}$$

$$= 7.20\text{N (left)}$$

$$+q = 11\text{cm}$$

$$r = 11 - 8 = 0.03\text{m}$$

$$F_2 = k \frac{(1 \times 10^{-6})(-2 \times 10^{-6})}{(0.03)^2}$$

$$= (9 \times 10^9) (2.22 \times 10^{-9})$$

$$= 20.0\text{N (right)}$$

$$F_{\text{net}} = 20 - 7.2 = 12.80\text{N right}$$

② b)  $x=11\text{cm}$ ,  $-2q=1\text{cm}$ ,  $+q=5$ ,  $+3q=8$ ,  $-q=14$   $k=9 \times 10^9$

$$q = 1\mu\text{C}$$

$$E = k \frac{q}{r^2}$$

$$-2q = 1\text{cm}$$

$$r = 0.1\text{m}$$

$$E_1 = k \frac{2 \times 10^{-6}}{(0.1)^2}$$

$$E_1 = 1800\text{N/C (left)}$$

$$+q = 5\text{cm}$$

$$r = 0.06\text{m}$$

$$E_2 = (9 \times 10^9) \frac{1 \times 10^{-6}}{0.0036}$$

$$E_2 = 2500\text{N/C (right)}$$

$$+3q = 8\text{cm}$$

$$r = 0.03\text{m}$$

$$E_3 = k \frac{(3 \times 10^{-6})}{0.0009}$$

$$E_3 = 30.000\text{N/C (right)}$$

$$-q = 14\text{cm}$$

$$r = 0.03\text{m}$$

$$E_4 = k \frac{(1 \times 10^{-6})}{0.0009}$$

$$E_4 = 10,000\text{N/C (left)}$$

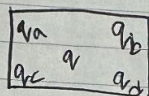
$$E_{\text{net}} = -1800 + 2500 + 30.000 - 10,000$$

$$E_{\text{net}} = 20700\text{N/C}$$

$$E_{\text{net}} = 2.07 \times 10^4\text{N/C}$$



2 a



$$q_a = q_b = +7.5 \mu\text{C} \quad \text{force} = ?$$

$$q_c = q_d = -7.5 \mu\text{C}$$

Since  $q_a = q_b$  &  $q_c = q_d$ , the net force on  $q = 0 \text{ N}$

b  $q_c = 2 \mu\text{C}$ ,  $l = 0.1 \text{ m}$

$$K = 9 \times 10^9$$

distance from point to  $q = r = \frac{s}{\sqrt{2}} = \frac{0.10}{\sqrt{2}} = 0.0707 \text{ m}$

$$F = k \frac{|q_c| |q|}{r^2} = k \frac{(2 \times 10^{-6})(7.5 \times 10^{-6})}{0.0707^2} = (9 \times 10^9)(3 \times 10^{-9}) = 27.0 \text{ N}$$

but since they are symmetrical, the force on  $q$  is still  $0 \text{ N}$

3 40 kV

non-relativistically find max speed

$$qV = \frac{1}{2}mv^2 \rightarrow v = \sqrt{\frac{2qV}{m}}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$V = 40,000 = 4 \times 10^4$$

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$v = \sqrt{\frac{2qV}{m}}$$

$$v = \sqrt{\frac{2(1.6 \times 10^{-19})(4 \times 10^4)}{9.11 \times 10^{-31}}}$$

$$v = 1.19 \times 10^8 \text{ m/s}$$

$$v = \sqrt{\frac{2(1.6 \times 10^{-19})(40000)}{9.11 \times 10^{-31}}} = 1.405 \times 10^8$$

$$v = 1.405 \times 10^8 \text{ m/s}$$



- ④ a) 4 cm separation is  $7.5 \times 10^4 \text{ Vm}^{-1}$   
what's  $\Delta V$  b/w plates?

$$\Delta V = E \cdot d = 7.5 \times 10^4 (0.04 \text{ m})$$

$$\Delta V = 3.0 \times 10^3 \text{ V}$$

- b) 1 cm from lower  
 $V = 7.5 \times 10^4 (0.01 \text{ m})$   
 $V = 750 \text{ V}$

c)  $E = \frac{\Delta V}{d}$   $\Delta V = 80 \text{ mV}$

$$d = 9 \text{ nm}$$

$$E = \frac{80 \times 10^{-3}}{9 \times 10^{-9}} = 8.89 \times 10^6 \text{ V/m}$$

- ⑤ double charged ion ( $E$ ) = 32.0 keV, separated by 2 cm

$$E = 32 \text{ keV} = 32 \times 10^3 \times 1.602 \times 10^{-19} \text{ J}$$

$$q = 2e = 2 \times 1.602 \times 10^{-19} \times 2 = 3.204 \times 10^{-19} \text{ C}$$

$$\Delta V = \frac{E}{q} = \frac{32 \times 10^3 \times 1.602 \times 10^{-19}}{3.204 \times 10^{-19}}$$

$$\Delta V = 16,000 \text{ V} \div 0.02 \text{ m} = 8.00 \times 10^5 \text{ V/m}$$

- ⑥ energy of  $\alpha$  nucleus was 5 MeV, how close to 79 protons before deflected?

$$E = 5 \text{ MeV} = 5 \times 10^6 \text{ eV}$$

$$Z_{\text{gold}} = 79 \text{ protons } Z_{\alpha} = 2$$

$$K = 8.99 \times 10^9 \text{ } e = 1.602 \times 10^{-19} \text{ C}$$

$$U(r) = \frac{K \cdot Z_{\text{gold}} \cdot Z_{\alpha} \cdot e^2}{r}$$

$$r = \frac{K \cdot Z_{\text{gold}} \cdot Z_{\alpha} \cdot e^2}{E}$$

$$r = \frac{(8.99 \times 10^9)(79)(2)(1.602 \times 10^{-19})^2}{(5 \times 10^6)(1.602 \times 10^{-19})} = \frac{(1.846 \times 10^{-28})}{8.01 \times 10^{-13}} = 2.3 \times 10^{-15} \text{ m, or } 2.3 \text{ femtometers}$$

## Unit 1

- ① capacitance to store  $3 \mu\text{C}$  at  $V = 120 \text{ V}$ ?

\*F = farad

$$C = \frac{Q}{V} = \frac{3 \times 10^{-6} \text{ C}}{120 \text{ V}} = 2.5 \times 10^{-8} \text{ F}$$



② a) energy = ?,  $10 \mu\text{F}$  capacitor to  $9 \times 10^3 \text{V}$

$$E = \frac{1}{2} CV^2$$

$$E = \frac{1}{2} (10 \times 10^{-6}) (9 \times 10^3)^2 = \frac{1}{2} (10 \times 10^{-6}) (8.1 \times 10^7)$$

$$E = 4.05 \times 10^2 \text{ J}$$

$$\boxed{E = 405 \text{ J}}$$

b)  $8 \mu\text{F}$  that stores  $40 \text{ J}$

$$E = \frac{1}{2} CV^2 \quad 8 \mu\text{F} = 8 \times 10^{-6} \text{ F}$$

$$V = \sqrt{\frac{2E}{C}}$$

$$V = \sqrt{\frac{2(40)}{8 \times 10^{-6}}}$$

$$V = \sqrt{\frac{80}{8 \times 10^{-6}}}$$

$$V = \sqrt{1 \times 10^7}$$

$$\boxed{V = 3162.3 \text{ V}}$$

③ a) 4 capacitors in parallel

$$E = 405 \text{ J} \quad V = 9 \times 10^3$$

$$C_{\text{tot}} = C_1 + C_2 + C_3 + C_4 = 4C$$

$$E = \frac{1}{2} (4C) V^2$$

$$405 = 2C (9 \times 10^3)^2$$

$$405 = 2C (8.1 \times 10^7)$$

$$405 = 1.62 \times 10^8 C$$

$$C = \frac{405}{1.62 \times 10^8}$$

$$= 2.5 \times 10^{-6} \text{ F} = 2.5 \mu\text{F}$$

$$\text{each capacitor: } Q = 2.5 \times 10^{-6} \times 9 \times 10^3$$

$$\boxed{Q = 22.5 \text{ mC}}$$

b) in series? why no?

$$\frac{1}{C_{\text{tot}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4}$$

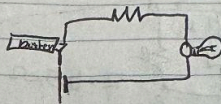
$$= \frac{1}{2.5} + \frac{1}{2.5} + \frac{1}{2.5} + \frac{1}{2.5} \text{ versus } 2.5 + 2.5 + 2.5 + 2.5$$

in parallel there's more charge, which is why the designer wants the capacitors in parallel.



④ 1k $\Omega$  resistor 3V battery

① flow from battery, is 3 $\Omega$  in the LED resistor?



$$R_{LED} = 3\Omega, R_{resist} = 1k\Omega = 1000\Omega, V_{bat} = 3V$$

$$R_{tot} = 1003\Omega$$

$$I = \frac{V}{R_{tot}} = \frac{3V}{1003\Omega} = 0.00299A = \boxed{2.99mA}$$

② power <sup>consumed</sup> by LED?

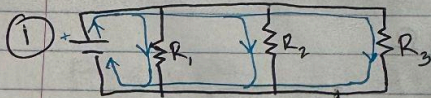
$$P = I^2 \cdot R_{LED}$$

$$P = (0.00299A)^2 \cdot 3\Omega$$

$$P = 8.94 \times 10^{-6} A (3\Omega)$$

$$\boxed{P = 2.66 \times 10^{-5} W = 26.6 \mu W}$$

## Unit 2



$$R_1 = 10k\Omega, R_2 = 5k\Omega, R_3 = ?$$

$$R_{tot} = 2k\Omega$$

$$= 2000\Omega$$

① resistance of  $R_3$ ?

$$\text{in parallel: } \frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{2000} = \frac{1}{10000} + \frac{1}{5000} + \frac{1}{R_3}$$

$$\frac{1}{2000} = \frac{3}{10000} + \frac{1}{R_3}$$

$$\frac{1}{R_3} = \left( \frac{1}{2000} \right)^{\times 5} - \frac{3}{10000}$$

$$\frac{1}{R_3} = \frac{5-3}{10000}$$

$$R_3 = \frac{10000}{2} = \boxed{5000\Omega} = \boxed{5k\Omega}$$

②  $R_{tot} = 2000\Omega, \Delta V = 12V$

$$I = \frac{V}{R_{tot}}$$

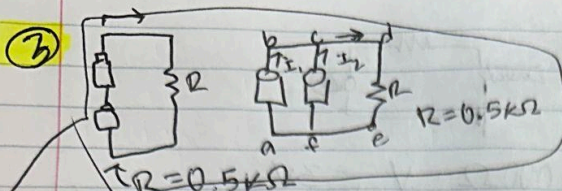
$$I = \frac{12V}{2000\Omega} = \boxed{0.006A} = \boxed{6mA}$$

$$\text{③ } I_1 = \frac{12V}{10000\Omega} = 0.0012A = \boxed{1.2mA}$$

$$I_2 = \frac{12V}{5000\Omega} = 0.0024A = \boxed{2.4mA}$$

$$I_3 = \frac{12V}{5000\Omega} = \boxed{2.4mA}$$





3 1.5V resistance of  $0.02\Omega$  in series  
w/ 1.53V carbon-zinc w/  
 $0.1\Omega$  resistance, load resistance  $10\Omega$

① current flows?

$$I = \frac{V}{R}$$

$$I = \frac{(3 \times 1.56V) + 1.53V}{(3 \times 0.02\Omega) + 10.1\Omega} = \frac{6.27V}{10.16\Omega} = \boxed{0.617A \text{ or } 617mA}$$

② power to load?

$$P = I^2 R = (.617)^2 \times 10\Omega = \boxed{3.8W}$$

③ dry cell = bad, only  $0.5W$  to load.

$$I = \sqrt{\frac{P}{R}} = \sqrt{\frac{0.5}{10}} = \sqrt{0.05} = 0.224A$$

$$R_{tot} = \frac{V_{tot}}{I} = \frac{6.27V}{0.224A} = 28.04\Omega$$

$$R_{internal} = R_{tot} - R_{load} = 28.04\Omega - 10\Omega = \boxed{18.04\Omega}$$

②  $R = 0.5k\Omega = 500\Omega$

①  $V_{series} = 1.5 + 1.5 = 3V$

$V_{parallel} = 3V$

Series:

$$I_{series} = \frac{V}{R} = \frac{3V}{500\Omega} = 0.006A = \boxed{6.0mA} \text{ (current)}$$

$$P_{series} = I^2 R = (.006)^2 (500) = 0.018W = \boxed{18mW} \text{ (power)}$$

Parallel:

$$I_{parallel} = \frac{3V}{500\Omega} = \boxed{6.0mA \text{ current}}$$

$$P_{parallel} = (.006)^2 (500) = \boxed{18mW \text{ power}}$$



(b)  $r = 0.5 \Omega$

Series

$V = 3V, R = 1 \Omega, R_{tot} = 501 \Omega$

$I = \frac{V}{R} = \frac{3V}{501 \Omega} = 5.99 \text{ mA}$  current

$P = I^2 R = 5.99^2 (500) = 17.94 \text{ mW}$  power

parallel

$0.5/2 = 0.25, R_{tot} = 500.25 \Omega$

$I = \frac{3V}{500.25} = 5.995 \text{ mA}$  current

$P = I^2 R = 5.995^2 (500) = 17.97 \text{ mW}$  power

### Unit 3

①

| case | v direction          | B direction          | F direction            |
|------|----------------------|----------------------|------------------------|
| (a)  | $\hat{i}$ , right    | $\hat{j}$ , up       | $-\hat{k}$ (into page) |
| (b)  | $\hat{j}$ , up       | $\hat{k}$ , out page | $-\hat{i}$ , left      |
| (c)  | $\hat{k}$ , out page | $\hat{i}$ , right    | $-\hat{j}$ , down      |

$F = q(v \times B)$

negative:  $q < 0$

(a)  $\hat{i} \times \hat{j} = \hat{k}$

$v$  is  $\hat{i}$  right  $\hat{j}$ ,  
 $B$  is  $\hat{j}$  up,  $q < 0$  so  $-\hat{k}$   
into page.

(b)  $\hat{j} \times \hat{k} = \hat{i}$

$v$  is  $\hat{j}$  up,  $\hat{k}$  is out,  
 $-\hat{i}$  is left

(c)  $\hat{k} \times \hat{i} = \hat{j}$

$\hat{k}$  is out page,  
 $\hat{i}$  is right,  $-\hat{j}$  is down

(2)  $4 \times 10^3 \text{ m/s} = v, 1.25 \text{ T} = B, F = 1.4 \times 10^{-16} \text{ N}, q = 1.6 \times 10^{-19} \text{ C}$

$F = qvB \sin \theta$

$\sin \theta = \frac{F}{qvB}$

$\sin \theta = \frac{1.4 \times 10^{-16}}{(1.6 \times 10^{-19})(4 \times 10^3)(1.25)} = \frac{1.4 \times 10^{-16}}{8 \times 10^{-16}} = 0.175$

Case A

$-\hat{j}$  (left)

$\theta = \arcsin(0.175)$

$\theta \approx 10.1^\circ$  or

$180 - 10.1 = 169.9^\circ$



- ③ (a)  $O^{-16}$  ion mass =  $2.66 \times 10^{-26} \text{ kg}$  @  $5 \times 10^6 \text{ m/s}$   
 perp to  $1.2 \text{ T}$ , circle arc  $0.231 \text{ m}$  radius  $r$ .  
 What positive charge is on ion?

$$qvB = \frac{mv^2}{r} \quad q = \frac{mv}{Br}$$

$$q = \frac{(2.66 \times 10^{-26})(5 \times 10^6)}{(1.2)(0.231)}$$

$$q = \frac{1.33 \times 10^{-19}}{0.2772} = \boxed{4.8 \times 10^{-19} \text{ C}}$$

- (b) ratio of charge to electron?

$$\frac{q}{e} = \frac{4.8 \times 10^{-19}}{1.6 \times 10^{-19}} = 3; \text{ 3x the charge of an electron}$$

- (c) radius for oxygen  $^{18}$ ?

Scaling, so:

$$\frac{r_{18}}{r_{16}} = \frac{m_{16}}{m_{18}} = 1.125$$

$$r_{18} = 1.125 \times r_{16}$$

$$r_{18} = 1.125 \times 0.231 = \boxed{0.26 \text{ m}}$$